Database Technologies - Assignment 9

Create appropriate indexes to solve the following problems.
Note: use “sales” database to solve below queries.

1. Create an index that will enable a user to pull orders for ‘1990-10-03’ out of the Orders table quickly.

```
mysql> CREATE INDEX spdate ON orders(odate);
Query OK, 0 rows affected (0.022 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM orders WHERE odate='1990-10-03';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
5 rows in set (0.002 sec)
```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

```
mysql> CREATE UNIQUE INDEX unicol on orders(onum);
Query OK, 0 rows affected (0.024 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM orders WHERE onum = '3001';
+------+-------+------------+------+------+
| onum | amt   | odate      | cnum | snum |
+------+-------+------------+------+------+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
+------+-------+------------+------+------+
1 row in set (0.001 sec)
```

3. Create an index that would permit salesperson to retrieve his orders.

```
mysql> CREATE INDEX sales_orders ON orders(snum);
Query OK, 0 rows affected (0.032 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM orders WHERE snum = '1001';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.001 sec)

mysql> SELECT * FROM orders WHERE snum = '1002';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3007 |   75.75 | 1990-10-04 | 2004 | 1002 |
| 3010 |  309.95 | 1990-10-04 | 2004 | 1002 |
+------+---------+------------+------+------+
3 rows in set (0.001 sec)
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

```
mysql> CREATE UNIQUE INDEX custo_rating ON customers(snum, rating);
ERROR 1062 (23000): Duplicate entry '1001-100' for key 'customers.custo_rating'
mysql>
```

5. Create an index to speed up searching order on a given date by given customer.

```
mysql> CREATE INDEX cus_order_date ON orders(cnum, odate);
Query OK, 0 rows affected (0.018 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM order WHERE cnum = '2001' AND odate = '1990-10-03';
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to yo
ur MySQL server version for the right syntax to use near 'order WHERE cnum = '2001' AND odate = '
1990-10-03'' at line 1
mysql> SELECT * FROM orders WHERE cnum = '2001' AND odate = '1990-10-03';
+------+--------+------------+------+------+
| onum | amt    | odate      | cnum | snum |
+------+--------+------------+------+------+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
+------+--------+------------+------+------+
1 row in set (0.003 sec)

mysql> SELECT * FROM orders WHERE cnum = '2004' AND odate = '1990-10-04';
+------+--------+------------+------+------+
| onum | amt    | odate      | cnum | snum |
+------+--------+------------+------+------+
| 3007 |  75.75 | 1990-10-04 | 2004 | 1002 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
+------+--------+------------+------+------+
2 rows in set (0.000 sec)
```